Assignment 6

lørdag 22. februar 2025 14:23

Problem 1

a) Ohm's law: $Z_L = \frac{V_L}{I_L}$ and $V_L = 132000V$

Complex power: $S_L = V_L \cdot \overline{I_L} \Rightarrow I_L = \overline{\frac{S_L}{V_L}}$

$S_L = 50 \angle 36,87^\circ \, MVA = 50 \cdot 10^6 \angle 36,87^\circ \, VA$

$\overline{S_L} = 50 \cdot 10^6 \angle -36,87^\circ$

$$I_L = \frac{50 \cdot 10^6 \angle -36,87^\circ \, VA}{132000 \, V} = 378,78 \angle -36,87^\circ A$$

$$Z_L = \frac{132000 \, V}{387,78 \angle -36,87^\circ A} = 348,4 \angle 36,87^\circ \, \Omega$$

Alternatively: $S_L = \frac{V_L^2}{Z_L} \Rightarrow Z_L = \frac{V_L^2}{S_L}$

Power factor: $pf = \cos(\varphi_{PF}) = \cos(36,87^\circ) = 0,8$

Positive pf => Current lags voltage

b) See task a)

c) KVL: $V_S = V_L + \underbrace{I_L \cdot Z_{Line}}_{V_{Line}}$

$Z_{Line} = R_{Line} + jX_{Line} = (10 + 35j)\Omega$

$Z_{Line} = 36,4 \angle 74,05^\circ$

$Z_{Line} = 36,4 \angle 74,05°$

$V_{Line} = I_L \cdot Z_{Line} = 378,79 \angle -36,87° \cdot 36,4 \angle 74,05° = 13788 \angle 36,18° \, V$

$V_S = V_L + V_{Line} = 132000 \angle 0° + 13788 \angle 36,18°$

Cartesian:

$V_{Line} = (11000 + 8320j) V$ and $V_L = 132000$

$V_S = 132000 + 11000 + 8320j = 143000 + 8320j$

$|V_S| = \sqrt{143000^2 + 8320^2} = 143241 \, V$

$\angle V_S = \tan^{-1}\left(\frac{8320}{143000}\right) = 3,33°$

$V_S = 143241 \angle 3,33° \, V$

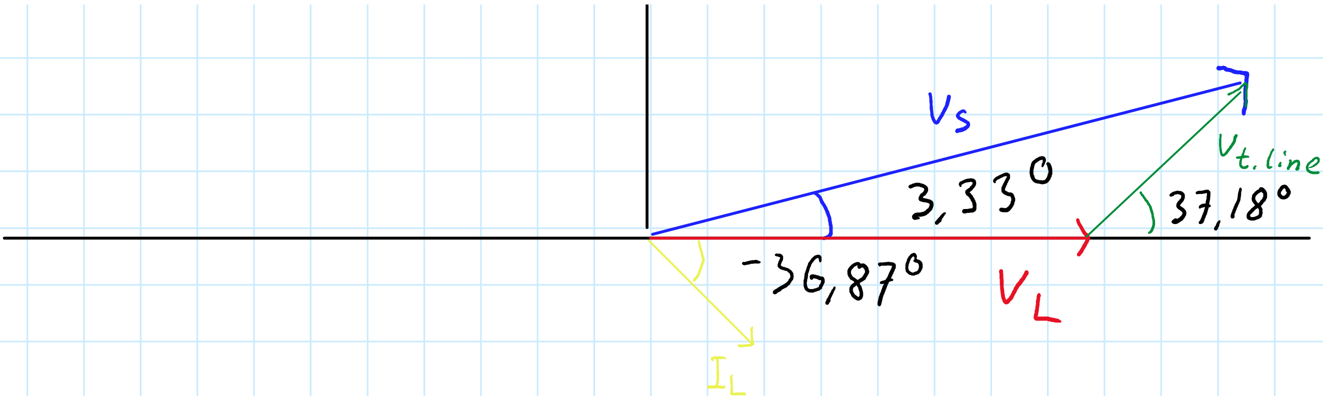
d) $S = V_S \cdot \overline{I_L} = 143230 \angle 3,33° \cdot 378,79 \angle 36,87°$

$= 54,25 \angle 40,2° \, MVA$

$P = |S| \cos(\varphi_{pf}) = 54,22 \cdot 10^6 \cos(40,2°) = 41,44 \, MW$

$Q = |S| \sin(\varphi_{pf}) = 54,22 \cdot 10^6 \sin(40,2°) = 35,02 \, MVAr$

e)



Problem 2

a) $V_L = V_S = 22000\,V$

b) $P = S_L \cos(\varphi) \Rightarrow |S_L| = \dfrac{P}{\cos\varphi} = \dfrac{5 \cdot 10^6}{0,9} = 5,556\,MVA$

$\varphi = \cos^{-1}(0,9) = 25,84°$

$S_L = 5,556\angle 25,84°\,MVA$

$Z_L = \dfrac{V_L^2}{\bar{S}}$, $\quad \bar{S} = 5,556\angle -25,84\,MVA$

$Z_L = \dfrac{(22000)^2}{5,556 \cdot 10^6 \angle -25,84°} = 87,05\angle 25,84°\,\Omega$

c) $|S| = \dfrac{P}{\cos\varphi} = 5,56\,MVA$

$I_L = \dfrac{|S|}{V_L} = \dfrac{5,56 \cdot 10^6}{22000} = 252,73\,A$ and $\varphi = 25,84°$

$I_L = 252,73\angle -25,84°\,A$

$V_{drop} = I_L \cdot jX_{line} = 252,73\angle -25,84° \cdot 10\angle 90°$

$= 2527,3\angle 64,16°$

$V_S = V_L + V_{drop} = 22000 + 2527,3\angle 64,16°$

$V_S = V_L + V_{drop} = 22000 + 2027,5\angle 64,16$

$|V_S| = 23219,2\ V$

Percentage drop: $\dfrac{23219,2 - 22000}{23219,2} \cdot 100 = 5,21\%$

d) Voltage drop: $V_{drop} = I_L \cdot jX_{Line}$

Reducing reactive power $\rightarrow$ current $\downarrow$ $\rightarrow$ Voltage $\downarrow$

Capacitor on land: Line still carries full reactive current

e) $Q_L = \sqrt{S^2 - P^2} = 2,43 \cdot 10^6\ VAR$

$Q_C = V_L^2 \cdot \omega C_L$ where $\omega = 2\pi f$ and $f = 50Hz$

$Q_C = 15,94\ \mu F$

f) $V_{drop} = I_L \cdot jX_{Line}$ where $X_{Line} = 10\ \Omega$

Current is real (no phase shift)

$I_L = \dfrac{P}{V_L} = \dfrac{5 \cdot 10^6}{22000} A = 227,27A$

$V_{drop} = 227,27 \cdot j10 \Rightarrow |V_{drop}| = 2272,7\ V$

$V_L = V_S - V_{drop} = 22000 - j2272,7$

$|V_L| = 22116,4V$

$\dfrac{V_S - V_L \cdot 100}{22000} = \dfrac{-116,4}{22000} = -0,53\%$

Problem 3

a) $V_{th} = 240V$

$V_{ab} = \dfrac{Z_L}{Z_L + R_{th} + jX_{th}} \Rightarrow V_{ab} = \dfrac{90 - 30j}{(90 + R_{th}) + (X_{th} - 30)j} V_{th}$

$$144\angle -36{,}86^\circ = \frac{94{,}87\angle -18{,}43}{r\angle\theta}\cdot 240$$

$$r\angle\theta = 158{,}117\angle 18{,}43$$

$R_{th} = 60$, $X_{th} = 80$

b) Skip

c) $Z_L = 60 - 80j$

$$S = \frac{240^2\cdot(60-80j)}{120^2} = 240 - 320j$$